

NE511 Computational Project Notes

Kyle Hansen

April 12, 2026

Contents

1	Multiphysics Problem Description	2
1.1	Boltzmann Transport Equation for multiplying medium with delayed neutrons . . .	2
1.1.1	Multigroup prompt neutron transport equation with Backward Euler time discretization	2
1.1.2	Precursor balance equations	4
1.1.3	Equivalent steady-state transport problem with delayed neutrons	4
1.1.4	Multigroup Low-Order Quasidiffusion Equations	5
1.1.5	Grey Low-Order Quasidiffusion Equations	6
1.1.6	Grey Coefficient Derivatives	7
1.2	Advection-diffusion equation	7
1.3	Cross-section feedback	8
2	MNP Method	8
2.1	Basic method details	9
2.2	Iterative Algorithms	9
3	NILO-MNP method	10
3.1	Method details	10
3.2	Iterative Algorithm	10
4	Reactor Calculation Algorithms	10
4.1	Criticality calculation	10
4.2	Criticality search	11
5	Benchmark Problems	11
5.1	PWR Control Rod Ejection [1]	11
6	Tests	11
6.1	Pure Absorber Transport	11
6.2	Pure Diffusion TH	11
6.3	Advection with Local Removal TH	11
Appendix A LD Discretization of Neutron Transport equation		16
Appendix B LD Discretization of QD equations		18

Appendix C LDFEM Discretization of advection-diffusion equation	22
C.1 Advective Interface Condition	24
C.2 Diffusive Interface Condition	24
Appendix D Computational Units	26
Appendix E List of Symbols	27

1 Multiphysics Problem Description

1.1 Boltzmann Transport Equation for multiplying medium with delayed neutrons

1.1.1 Multigroup prompt neutron transport equation with Backward Euler time discretization

The problem described here is for 1-dimensional multigroup neutron transport, coupled to 1-dimensional advection-diffusion through temperature- and flux-dependent cross sections.

The one-dimensional time-dependent neutron transport equation (NTE) for prompt neutrons with isotropic scattering, isotropic source, and vacuum boundary conditions is

$$\left[\frac{1}{v(E)} \frac{\partial}{\partial t} + \mu \frac{\partial}{\partial x} + \sigma_t(x, E) \right] \psi(x, \mu, E, t) = \frac{\chi(E)}{2} \int_0^\infty dE' (1 - \beta(E')) \nu(x, E') \sigma_f(x, E') \phi(x, E', t) + \frac{1}{2} \int_0^\infty dE' \sigma_s(x, E' \rightarrow E) \phi(x, E', t) + q(x, E, t),$$

$$x \in [0, X], E \in [0, \infty), \mu \in [-1, 1], t > 0 \quad (1a)$$

$$\phi(x, E, t) = \int_{-1}^1 d\mu \psi(x, \mu, E, t) \quad (1b)$$

$$\psi(x, \mu, E, 0) = \psi^0(x, \mu, E) \quad (1c)$$

$$\psi(0, \mu, E, t) = 0, \quad \mu > 0 \quad (1d)$$

$$\psi(X, \mu, E, t) = 0, \quad \mu < 0. \quad (1e)$$

Using the standard multigroup approximation, the equation for group g becomes

$$\left[\frac{1}{v_g} \frac{\partial}{\partial t} + \mu \frac{\partial}{\partial x} + \sigma_{t,g} \right] \psi_g(x, \mu, E, t) = \frac{\chi_g}{2} \sum_{g'=1}^{N_g} (1 - \beta_{g'}) \nu \sigma_{f,g'}(x) \phi_{g'}(x, t) + \frac{1}{2} \sum_{g'=1}^{N_g} \sigma_{s,g' \rightarrow g}(x) \phi_{g'}(x, t) + q_g(x, t),$$

$$g = 1, \dots, N_g, x \in [0, X], \mu \in [-1, 1], t > 0 \quad (2a)$$

$$\phi_g(x, t) = \int_{-1}^1 d\mu \psi_g(x, \mu, t) \quad (2b)$$

$$\psi_g(x, \mu, 0) = \psi_g^0(x, \mu) \quad (2c)$$

$$\psi_g(0, \mu, t) = 0, \quad \mu > 0 \quad (2d)$$

$$\psi_g(X, \mu, t) = 0, \quad \mu < 0. \quad (2e)$$

The Backward Euler temporal discretization scheme is

$$\left. \frac{\partial \psi(t)}{\partial t} \right|_{t=t^j} \approx \frac{\psi(t^j) - \psi(t^{j-1})}{\Delta t_j} \quad (3)$$

where $t^{j=0} = 0$ and

$$\Delta t_j = t^j - t^{j-1}. \quad (4)$$

Substituting this into the multigroup neutron transport equation yields the effective steady-state problem at each time step,

$$\begin{aligned} \left[\mu \frac{\partial}{\partial x} + \hat{\sigma}_{t,g}^{(j)} \right] \psi_g^{(j)}(x, \mu, E) = \\ \frac{\chi_g}{2} \sum_{g'=1}^{N_g} (1 - \beta_{g'}) \nu \sigma_{f,g'}^{(j)}(x) \phi_{g'}^{(j)}(x) + \frac{1}{2} \sum_{g'=1}^{N_g} \sigma_{s,g' \rightarrow g}^{(j)}(x) \phi_{g'}^{(j)}(x) + \hat{q}_g^{(j)}(x, \mu), \\ g = 1, \dots, N_g, \quad j = 1, \dots, N_j, \quad x \in [0, X], \quad \mu \in [-1, 1] \end{aligned} \quad (5a)$$

$$\phi_g^{(j)}(x) = \int_{-1}^1 d\mu \psi_g^{(j)}(x, \mu) \quad (5b)$$

$$\psi_g^{(0)}(x, \mu) = \psi_g^0(x, \mu) \quad (5c)$$

$$\psi_g^{(j)}(0, \mu) = 0, \quad \mu > 0 \quad (5d)$$

$$\psi_g^{(j)}(X, \mu) = 0, \quad \mu < 0 \quad (5e)$$

where the augmented total cross section and source are

$$\hat{\sigma}_{t,g}^{(j)} = \left(\sigma_{t,g}^{(j)}(x) + \frac{1}{v_g \Delta t_j} \right) \quad (6)$$

$$\hat{q}_g^{(j)}(x, \mu) = \left(q_g(x, t_j) + \frac{1}{v_g \Delta t_j} \psi_g^{(j-1)}(x, \mu) \right). \quad (7)$$

1.1.2 Precursor balance equations

In one dimension, the precursor balance equation is

$$\frac{\partial}{\partial t} C_{d,k}(x, t) = -\lambda_k C_{d,k}(x, t) + \sum_{g=1}^{N_g} \beta_{k,g}(x, t) \nu \sigma_{f,g}(x) \phi_g(x, t), \quad k = 1, \dots, 6, \quad t > 0 \quad (8a)$$

$$C_{d,k}(x, 0) = C_{d,k}^0(x). \quad (8b)$$

With the Backward Euler temporal discretization, this may be written, after some algebra, as

$$C_{d,k}^{(j)}(x) = \frac{\frac{C_{d,k}^{(j-1)}}{\Delta t_j} + \sum_{g=1}^{N_g} \beta_{g,k} \nu \sigma_{f,g} \phi_g^{(j)}}{\lambda_k + \frac{1}{\Delta t_j}}. \quad (9)$$

In the neutron transport equation, the delayed neutron source is

$$q_{del,g}(x, \mu) = \frac{1}{2} \sum_{k=1}^6 \lambda_k \chi_{g,k}^d C_{d,k}, \quad (10)$$

so, substituting eq. (9) into eq. (10), the time-discretized delayed neutron source in the transport equation becomes

$$q_{del,g}(x, \mu) = \frac{1}{2} \sum_{g'=1}^{N_g} \Gamma_{g' \rightarrow g}^{(j)} \nu \sigma_{f,g'} \phi_{g'}^{(j)} + q_{del,g}^{\chi,(j)}(x, \mu) \quad (11)$$

where

$$\Gamma_{g' \rightarrow g}^{(j)} = \sum_{k=1}^6 \chi_{g,k} \frac{\lambda_k \Delta t_j \beta_{g',k}}{\lambda_k \Delta t_j + 1} \quad (12)$$

$$q_{del,g}^{\chi,(j)}(x, \mu) = \frac{1}{2} \sum_{k=1}^6 \lambda_k \chi_{g,k} \left(\frac{C_{d,k}^{(j-1)}}{\lambda_k \Delta t_j + 1} \right). \quad (13)$$

Here, $q_{del}^{\chi,(j)}$ represents the decay of precursors that existed in time step j , and $\Gamma_{g' \rightarrow g}^{(j)}$ represents the proportion of fissions in group g' , during the time step Δt_j , whose delayed neutrons decay during the same time step (immediate precursor decay).

1.1.3 Equivalent steady-state transport problem with delayed neutrons

Adding eq. (11) to eq. (5), the quasi-steady state transport equation to be solved at each time step, including the effect of delayed neutrons, is

$$\begin{aligned}
\left[\mu \frac{\partial}{\partial x} + \hat{\sigma}_{t,g}^{(j)} \right] \psi_g^{(j)}(x, \mu, E) = \\
\frac{\chi_g}{2} \sum_{g'=1}^{N_g} (1 - \beta_{g'}) \nu \sigma_{f,g'}^{(j)}(x) \phi_{g'}^{(j)}(x) + \frac{1}{2} \sum_{g'=1}^{N_g} \left(\sigma_{s,g' \rightarrow g}^{(j)}(x) + \Gamma_{g' \rightarrow g}^{(j)} \nu \sigma_{f,g'} \right) \phi_{g'}^{(j+1)}(x) \\
+ \hat{q}_g^{(j)}(x, \mu) + q_{del,g}^{\chi,(j)}(x, \mu), \\
g = 1, \dots, N_g, \quad j = 1, \dots, N_j, \quad x \in [0, X], \quad \mu \in [-1, 1] \quad (14a)
\end{aligned}$$

$$\phi_g^{(j)}(x) = \int_{-1}^1 d\mu \psi_g^{(j+1)}(x, \mu) \quad (14b)$$

$$\psi_g^{(j)}(0, \mu) = 0, \quad \mu > 0 \quad (14c)$$

$$\psi_g^{(j)}(X, \mu) = 0, \quad \mu < 0 \quad (14d)$$

where

$$\psi_g^{(0)}(x, \mu) = \psi_g^0(x, \mu), \quad g = 1, \dots, N_g \quad (15a)$$

$$C_{d,k}^{(0)}(x) = C_{d,k}^0(x), \quad k = 1, \dots, 6 \quad (15b)$$

After solving for the neutron distribution, the precursor distribution can be recovered using eq. (9).

1.1.4 Multigroup Low-Order Quasidiffusion Equations

Taking the zeroth and first spatial moments ($\int \cdot d\mu$ and $\int \mu \cdot d\mu$) of eq. (14) (omitting $j+1$ index) yields

$$\begin{aligned}
\frac{\partial}{\partial x} J_g(x) + \sigma_{t,g} \phi_g(x) = \chi_g \overline{(1 - \beta) \nu \sigma_f \phi} \\
+ \sum_{g'=1}^{N_g} \left(\sigma_{s,g' \rightarrow g} + \Gamma_{g' \rightarrow g} \nu \sigma_{f,g'} \right) \phi_{g'}(x) + \hat{Q}_g(x) + Q_{del,g}^{\chi} \quad (16a)
\end{aligned}$$

$$\frac{\partial}{\partial x} (E_g \phi_g) + \sigma_{t,g} J_g = \tilde{q}_g \quad (16b)$$

$$J_g(0) = C_{0,g}^b \phi_g(0), \quad J_g(X) = C_{X,g}^b \phi_g(X) \quad (16c)$$

and the initial conditions are derived from taking the zeroth and first spatial moments of ψ^0 and

$$\hat{Q}_g(x) = \left(2q_g(x) + \frac{1}{v_g \Delta t_j} \phi_g^{(j)}(x) \right) \quad (17)$$

$$Q_{del,g}^{\chi} = \sum_{k=1}^6 \lambda_k \chi_{g,k} \left(\frac{C_k^{(j)}}{\lambda_k \Delta t_j + 1} \right) \quad (18)$$

$$\tilde{q}_g = \frac{1}{v_g \Delta t_j} J_g^{(j)}(x) \quad (19)$$

and

$$\overline{(1 - \beta)\nu\sigma_f} = \frac{\sum_{g=1}^{N_g} (1 - \beta_g)\nu\sigma_{f,g}}{\sum_{g=1}^{N_g} \phi_g} \quad (20)$$

$$\phi = \sum_{g=1}^{N_g} \phi_g. \quad (21)$$

The nonlinear closure term (Eddington Factor) is

$$E_g = \frac{\int_{-1}^1 \mu^2 \psi_g d\mu}{\int_{-1}^1 \psi_g d\mu} \quad (22)$$

and the boundary factors are

$$C_{0,g}^b = \frac{\int_{-1}^0 \mu \psi_g(0, \mu) d\mu}{\int_{-1}^0 \psi_g(0, \mu) d\mu} \quad (23)$$

$$C_{X,g}^b = \frac{\int_0^1 \mu \psi_g(X, \mu) d\mu}{\int_0^1 \psi_g(X, \mu) d\mu}. \quad (24)$$

1.1.5 Grey Low-Order Quasidiffusion Equations

Summing Equation (16) over all energy groups yields

$$\frac{\partial J}{\partial x} + \left(\bar{\sigma}_a - \overline{(1 - \beta)\nu\sigma_f} \right) \phi = \bar{Q} + \bar{Q}_{del}^x \quad (25a)$$

$$\frac{\partial}{\partial x} (\bar{E}\phi) + \bar{\sigma}_{tr} J + \bar{\zeta}\phi = \bar{q} \quad (25b)$$

$$J(0) = \bar{C}_0^b \phi(0), \quad J(X) = \bar{C}_X^b \phi(X) \quad (25c)$$

where

$$\bar{\sigma}_a = \frac{\sum_{g'} \left[\sigma_{t,g'} - \sum_g (\sigma_{s,g' \rightarrow g} + \Gamma_{g' \rightarrow g} \nu \sigma_{f,g'}) \right] \phi_{g'}}{\sum_{g'} \phi_{g'}} \quad (26)$$

$$\bar{Q} = \sum_g \hat{Q}_g, \quad \bar{Q}_{del}^x = \sum_g Q_{del,g}^x \quad (27)$$

$$\bar{E} = \frac{\sum_g E_g \phi_g}{\sum_g \phi_g} \quad (28)$$

$$\bar{\sigma}_{tr} = \frac{\sum_g \sigma_{t,g} |J_g|}{\sum_g |J_g|} \quad (29)$$

$$\bar{q} = \sum_g \tilde{q}_g \quad (30)$$

and $\bar{\zeta}\phi$ is a correction term accounting for the absolute value of current introduced in $\bar{\sigma}_{tr}$,

$$\bar{\zeta} = \frac{\sum_g (\sigma_{t,g} - \bar{\sigma}_{tr}) J_g}{\sum_g \phi_g}. \quad (31)$$

1.1.6 Grey Coefficient Derivatives

The grey coefficients $\bar{\sigma}_a$, $\overline{(1-\beta)\nu\sigma_f}$, $\bar{\sigma}_{tr}$, and $\bar{\zeta}$ are all temperature- and power-dependent. Then their temperature derivatives, needed for the NILO method, are

$$\dot{\bar{\sigma}}_{a,T} = \frac{1}{\sum_g \phi_g} \sum_{g'=1}^{N_g} \phi_{g'} \left(\frac{\partial}{\partial T} \sigma_{t,g'} + \sum_g \left(\frac{\partial}{\partial T} \sigma_{g' \rightarrow g} + \Gamma_{g' \rightarrow g} \frac{\partial}{\partial T} \nu \sigma_{f,g'} \right) \right) \quad (32a)$$

$$\overline{\dot{(1-\beta)\nu\sigma_{fT}}} = \frac{1}{\sum_g \phi_g} \sum_g \phi_g (1-\beta_g) \frac{\partial}{\partial T} \nu \sigma_{f,g} \quad (32b)$$

$$\dot{\bar{\sigma}}_{tr,T} = \frac{1}{\sum_g |J_g|} \sum_g |J_g| \frac{\partial}{\partial T} \sigma_{t,g} \quad (32c)$$

$$\dot{\bar{\zeta}}_T = \frac{1}{\sum_g \phi_g} \sum_g J_g \left(\frac{\partial}{\partial T} \sigma_{t,g} - \dot{\bar{\sigma}}_{tr} \right) \quad (32d)$$

The power derivatives $\dot{\bar{\sigma}}_F = \frac{\partial}{\partial f} \bar{\sigma}$ are defined in the same way, ignoring weight-function (ϕ_g , $|J_g|$) dependence on reaction rate (since only an approximate derivative is needed), except for the average prompt fission cross section, which is strongly dependent on fission rate,

$$\overline{\dot{(1-\beta)\nu\sigma_{fF}}} = \frac{1}{\sum_g \phi_g}. \quad (33)$$

Note that the derivatives of augmented total cross section are equal to the derivatives of the unaltered total cross section.

1.2 Advection-diffusion equation

The advection-diffusion equation in 1D is

$$\left[\frac{\partial}{\partial t} - \frac{\partial}{\partial x} k_b \frac{\partial}{\partial x} + \dot{m} c_p \frac{\partial}{\partial x} \right] T(x, t) = W_f f(x, t) \quad (34)$$

or in operator form,

$$\mathcal{H}T(x, t) = Q_f \quad (35)$$

with appropriate boundary conditions. the fission reaction rate f is

$$f(x) = \sum_{g=1}^{N_g} \sigma_{f,g} \phi_g(x, t) \quad (36)$$

and

$$\begin{aligned} c_p &= \text{specific heat capacity} \\ k_b &= \text{turbulent diffusion coefficient} \\ \dot{m} &= \text{mass flow rate} \\ W_f &= \text{fission energy deposition} \end{aligned}$$

Discretizing with Backward Euler, this equation becomes

$$\left[\frac{1}{\Delta t_j} - \frac{\partial}{\partial x} k_b \frac{\partial}{\partial x} + \dot{m} c_p \frac{\partial}{\partial x} \right] T^{(j+1)}(x) = W_f f^{(j+1)}(x) + \frac{1}{\Delta t_j} T^{(j)} \quad (37)$$

$$\hat{\mathcal{H}}T^{(j+1)}(x) = \hat{Q}_f^{(j+1)} \quad (38)$$

which is an equivalent steady-state advection-diffusion problem at each time step with a local removal term.

1.3 Cross-section feedback

For this study, the macroscopic feedback will be modeled with

$$\sigma_u(x) = \sigma_{u,0} + \sigma_{u,1,d} \left(\sqrt{T(x) + \kappa W_f f(x)} - \sqrt{T_{d,0}} \right) + \sigma_{u,1,b} (T(x) - T_0) \quad (39)$$

for each reaction u , where $\kappa, \sigma_{u,0}, \sigma_{u,1,d}, \sigma_{u,1,b}, T_{d,0}$, and T_0 are material constants, defined for each neutron energy group g .

2 MNP Method

The MNP method [2] is a method for efficient coupling of multigroup neutron transport with coupling and precursor equations. It is based on the Multilevel nonlinear Quasidiffusion (QD) equations for solving the neutron transport equation in single-physics.

The method makes no physical approximation and its use of an exact projective operator makes even the innermost level equivalent to solving the high-order transport equation. It effectively reduces the high ($x \times \mu \times E$) dimensionality of the transport problem to the dimensionality of the TH operator (x only). The method accelerates transport iterations and converges faster than the FPI-QD method.

2.1 Basic method details

The FPI-QD method is similar to the FPI-CMFD method described in [3, 4], where each outer iteration consists of a cross section evaluation, solution of the transport equation accelerated by QD, then TH solution to yield T .

This FPI-QD method converges at the same rate as any other FPI multiphysics method (since the outer iterations are equivalent), though the solution of the transport equation is accelerated. FPI may be unstable and is slow to converge [3].

The MNP method instead couples the TH equation with the GLOQD equations. In each outer iteration, the high-order transport problem is used to form a projective operator which closes the MLOQD equations, whose solution is used to form the GLOQD equations. The coupled system of GLOQD, PCB, and HT equations is then solved with Newton iterations [5].

On each Newton iteration, the iterate is updated with

$$\mathbf{x}_{p+1} = \mathbf{x}_p - \mathbf{J}_F(\mathbf{x}_p)^{-1} F(\mathbf{x}_p) \quad (40)$$

where F represents the residual operator of all coupled equations, $\mathbf{J}_F$ is the Jacobian which may be explicitly formed as in [2] or replaced with a JFNK approach, and

$$\mathbf{x} = \begin{bmatrix} \phi(x) \\ J(x) \\ \bar{\sigma}_a(x) \\ (1 - \beta)\nu\sigma_f(x) \\ \bar{\sigma}_{tr}(x) \\ \bar{\zeta}(x) \\ T(x) \end{bmatrix} \quad (41)$$

2.2 Iterative Algorithms

The iterative algorithm for FPI-QD is shown in and the algorithm for MNP is shown in

Algorithm 1: QD Fixed-Point Iteration Multiphysics

```

Input: f
while TODO Temperature is not converged do
  while Scalar flux is not converged do
    while TODO Angular flux is not converged do
      | Sweep Transport;
    end
    while TODO Scalar Flux is not converged do
      | Solve MLOQD Equations;
      | Generate Grey coefficients;
      | Solve GLOQD Equations;
    end
  end
  Evalute TH Equations;
end

```

3 NILO-MNP method

3.1 Method details

3.2 Iterative Algorithm

4 Reactor Calculation Algorithms

4.1 Criticality calculation

Two types of criticality calculation can be performed, one with fixed temperature/cross sections (for calculating k_{eff} at a single time step during a transient), or one with a power normalization (for criticality search, etc). The first is performed by solving a steady-state problem, redefining the fission source as

$$q_{f,g}(x) = \chi_g^{\text{tot}}(x) \frac{1}{2} \sum_{g'=1}^{N_g} \frac{1}{k_{\text{eff}}} \nu \sigma_{f,g'}(x) \phi_{g'}(x) \quad (42)$$

where the fission emission spectrum is

$$\chi_g^{\text{tot}}(x) = (1 - \beta) \chi_g + \sum_{k=1}^6 \beta_{k,g} \chi_{g,k}^d. \quad (43)$$

Given fixed cross sections, the transport equation alone (without TH or precursor balance) is iterated to find k_{eff} which satisfies the steady-state problem. For the second type of criticality calculation, eqs. (42) and (43) are solved, coupled to TH and temperature-dependent cross sections, with an additional equation for power normalization,

$$P = \int_0^X dx W_f \sum_{g=1}^{N_g} \sigma_{f,g}(x) \phi_g(x) \quad (44)$$

again using χ^{tot} , without considering the precursor balance equations. Here P is a known power in Watts (e.g. nameplate power, shutdown power).

The first type of calculation (fixed cross-section) is performed in the typical way. The power-normalized type of calculation modifies the outer (n) iteration of whichever solution method is used. Before evaluating cross sections, k_{eff} is updated, then the scalar flux solution is scaled to satisfy eq. (44). The eigenvalue update is

$$k_{\text{eff}}^{(n)} = \frac{\int_0^X dx \sum_{g=1}^{N_g} \sigma_{f,g}^{(n)}(x) \phi_g^{(n)}(x)}{\int_0^X dx \sum_{g=1}^{N_g} \sigma_{f,g}^{(n-1)}(x) \phi_g^{(n-1)}(x)}. \quad (45)$$

Note that if W_f is space-independent, the power normalization and k_{eff} update can be grouped into the same step.

4.2 Criticality search

A criticality search is used to find the value of some parameter y that satisfies

$$k_{\text{eff}}(y, P) - k_{\text{eff}}^{\text{target}} = 0. \quad (46)$$

The parameter y could represent poison concentration, control rod position, inlet temperature, etc. The search is performed using a root-finding algorithm (bisection, Newton's method, etc) combined with power-normalized criticality calculations.

5 Benchmark Problems

5.1 PWR Control Rod Ejection [1]

Two-group problem with six delayed neutron groups;

The active height of the core is 250 cm, with 5 regions defined in Table 1 and constants defined in table 2.

The self-scattering and fission cross sections are independent of temperature, with absorption cross section governed by the equations

$$\sigma_{a,1} = \sigma_{a,1}^0 + 2.7 \times 10^{-5}(T - 878 \text{ K}) \quad (47)$$

$$\sigma_{s,1 \rightarrow 2} = \sigma_{s,1 \rightarrow 2}^0 \left(1 - \frac{2.7 \times 10^{-5}}{\sigma_{s,1 \rightarrow 2}^0}(T - 878 \text{ K}) \right). \quad (48)$$

A reflector is added on the top and bottom (to replace the albedo boundary conditions given in [1]).

Two versions of the transient are simulated. First, as described in [1],

1. Begin : steady-state conditions, bottom of rod x_r cm from top of core
2. $t = 0$ Rod begins ejection at constant velocity
3. $t = 0.05$ Rod stops travel, bottom of rod 0 cm from top of core
4. $t = 1.5$ Scram rod begins travel from 0.0 cm, at 40 cm/s
5. $t = 7.75$ Scram rod stops travel, fully inserted into core
6. End : continue simulation until $t = 15$.

In the second scenario, where the scram rod does not drop, only the first three steps are considered, and simulation continues for 15 seconds.

6 Tests

6.1 Pure Absorber Transport

6.2 Pure Diffusion TH

6.3 Advection with Local Removal TH

The equation for pure advection with general (space-dependent) coefficients is

$$rT(x) + \dot{m}c_p \frac{\partial}{\partial x} T(x) = Q, \quad x \in [0, X] \quad (49a)$$

$$T(0) = T^{in} \quad (49b)$$

or

$$T(x) + \frac{\dot{m}c_p}{r} \frac{\partial}{\partial x} T(x) = \frac{Q(x)}{r}. \quad (50)$$

Using an integrating factor, the analytical solution is finally

$$T(x) = T^{in} \exp \left[- \int_0^x \frac{r(x')}{\dot{m}c_p} dx' \right] + \int_0^x \frac{Q(x'')}{\dot{m}c_p} \exp \left[- \int_0^{x''} \frac{r(x')}{\dot{m}c_p} dx' \right] dx''. \quad (51)$$

Algorithm 2: Multilevel Nonlinear Projective Method

Input: $\psi_{g,m}^{(0)}, T^{(0)}, C_k^{(0)}$
 $j = 0;$
while $t^j \leq t^{end}$ **do**
 $j \leftarrow j + 1;$
 $\Gamma_{g' \rightarrow g}^j, q_{del,g}^{\chi,j}, Q_{del,g}^{\chi,j} \leftarrow$ implicit precursor terms using $C_k^{j-1};$
 $\hat{q}_g^j, \hat{Q}_g^j, \hat{q}_g \leftarrow$ augmented sources from temporal discretization;
 $n = 0;$
 $\phi^{j,(0)} \leftarrow \phi^{j-1}; T^{j,(0)} \leftarrow T^{j-1}; \sigma_{u,g}^{j,(0)} \leftarrow \sigma_{u,g}^{j-1};$
 Initialize closures from previous time step;
 while $\|1 - \phi^{j,(n)} / \phi^{j,(n-1)}\|_\infty > \varepsilon_{\phi,n}$ **or** $\|1 - T^{j,(n)} / T^{j,(n-1)}\|_\infty > \varepsilon_{T,n}$ **do**
 $n \leftarrow n + 1;$
 if $n > 1$ **then**
 $\sigma_{u,g}^{j,(n-1)} = \sigma_{u,g}(T^{j,(n-1)}, f^{j,(n-1)}); \hat{\sigma}_{t,g}^{j,(n-1)} = \sigma_{t,g}^{j,(n-1)} + \frac{1}{v_g \Delta t_j};$
 for $g = 1, \dots, N_g$ **do**
 $\psi_g^{j,(n)} \leftarrow$ high-order transport equation using $\phi_g^{j,(n-1)};$
 end
 $\overline{E}_g^{j,(n)}, E_g^{b;j,(n)}, C_g^{\pm;j,(n)}, G_g^{j,(n)} \leftarrow$ Closure equations using $\psi_g^{j,(n)};$
 end
 $s = 0;$
 $\phi^{j,(n,0)} \leftarrow \sum_g \phi_g^{j,(n)};$
 while $\|1 - \phi^{j,(n,s)} / \phi^{j,(n,s-1)}\|_\infty > \varepsilon_{\phi,s}$ **or** $\|1 - T^{j,(n,s)} / T^{j,(n,s-1)}\|_\infty > \varepsilon_{T,s}$ **do**
 $s \leftarrow s + 1;$
 $\sigma_{u,g}^{j,(n,s-1)} = \sigma_{u,g}(T^{j,(n,s-1)}, f^{j,(n,s-1)});$
 for $g = 1, \dots, N_g$ **do**
 $\phi_g^{j,(n,s)}, J_g^{j,(n,s)} \leftarrow$ MLOQD Equations using $\sigma_{u,g}^{j,(n,s-1)}, \overline{E}_g^{j,(n)}, E_g^{b;j,(n)}, C_g^{\pm;j,(n)}, G_g^{j,(n)};$
 end
 Calculate Grey coefficients;
 $l = 0;$
 $\phi^{j,(n,s,0)} \leftarrow \sum_g \phi_g^{j,(n,s)};$
 while $\|1 - \phi^{j,(n,s,l)} / \phi^{j,(n,s,l-1)}\|_\infty > \varepsilon_{\phi,l}$ **or** $\|1 - T^{j,(n,s)} / T^{j,(n,s-1)}\|_\infty > \varepsilon_{T,l}$ **do**
 $l \leftarrow l + 1;$
 Newton Iterations;
 Form Jacobian;
 Solve linear system;
 end
 $\phi_g^{j,(n,s)} \leftarrow (\phi_g^{j,(n,s,l)} / \sum_g \phi_g^{j,(n,s,l)}) \phi^{j,(n,s,l)};$
 $T^{j,(n,s)} \leftarrow T^{j,(n,s,l)};$
 end
 $\phi_g^{j,(n)} \leftarrow (\phi_g^{j,(n,s)} / \sum_g \phi_g^{j,(n,s)}) \phi^{j,(n,s)};$
 $T^{j,(n)} \leftarrow T^{j,(n,s)};$
 end
 $C_{d,k}^j \leftarrow$ Analytic precursor recovery using $\phi_g^j;$
end

Table 1: Zone Layout and materials

Location	Core Bottom						Core top
Material ID	R	M1	M2	M3	M2	M1	R
Zone height	50 cm	50 cm	50 cm	50 cm	50 cm	50 cm	50 cm

Table 2: Unrodded material cross sections [cm^{-1}]

M1 Group	1	2
σ_a^0	0.07582	0.01043
$\nu\sigma_f$	0.11120	0.004695
$\sigma_{g \rightarrow g}$	0.24015	0.91424
$\sigma_{g \rightarrow g+1}^0$	0.01536	–
M2 Group	1	2
σ_a^0	0.07681	0.01045
$\nu\sigma_f$	0.11130	0.004595
$\sigma_{g \rightarrow g}$	0.23964	0.91701
$\sigma_{g \rightarrow g+1}^0$	0.01521	–
M3 Group	1	2
σ_a^0	0.07691	0.01054
$\nu\sigma_f$	0.11140	0.004494
$\sigma_{g \rightarrow g}$	0.23946	0.91726
$\sigma_{g \rightarrow g+1}^0$	0.01520	–

Table 3: Graphite reflector cross sections [cm^{-1}]

Group	1	2
$\sigma_{a,g}$	0.0007	0.0003
$\sigma_{s,g \rightarrow g}$	0.29	0.40
$\sigma_{s,1 \rightarrow 2}$	0.0003	–

Table 4: Energy Production and Deposition

Thermal power	$1.531 \times 10^{-3} \text{W cm}^{-2}$
Energy deposition per fission	$330 \times 10^{-13} \text{ J}$
v_1	$2.0 \times 10^9 \text{ cm s}^{-1}$
v_2	$2.2 \times 10^5 \text{ cm s}^{-1}$

Table 5: Delayed Neutron data

k	decay constant λ_k [s ⁻¹]	β_k
1	0.0124	0.000215
2	0.0305	0.001424
3	0.1110	0.001274
4	0.3010	0.002568
5	1.1300	0.000748
6	3.0000	0.000273
total		0.006502

Table 6: Control Rod Cross section

Group	1	2
$\sigma_{a,g}$	$0.006769f$	$0.02583f$

Table 7: Thermal-hydraulic parameters

k_b	$1.5 \text{ W cm}^{-1} \text{ K}^{-1}$
$\dot{m}$	$0.2382 \text{ kg s}^{-1} \text{ cm}^{-2}$
T_{in}	505 K
c_p	$4670 \text{ J kg}^{-1} \text{ K}^{-1}$

References

- [1] A. M. Larsen, E. Nonbøl, and B. Thorlaksen, “The danish solution of the NRF benchmark problem 1980: Control rod ejection in a PWR,” 1981.
- [2] A. Tamang and D. Y. Anistratov, “A multilevel projective method for solving the space-time multigroup neutron kinetics equations coupled with the heat transfer equation,” *Nuclear Science and Engineering*, vol. 177, no. 1, pp. 1–18, 2014.
- [3] “The NILO-CMFD method for iteratively solving multiphysics neutron transport - thermal hydraulic problems,” in *Proceedings of the American Nuclear Society Conference on Mathematics and Computation 2021*, pp. 2092–2101, ANS, 2021.
- [4] N. J. Adamowicz, A. Manera, and E. W. Larsen, “The NILO-CMFD method for iteratively solving coupled neutron transport–thermal hydraulics problems,” *Nuclear Science and Engineering*, vol. 197, no. 2, pp. 262–278, 2023.
- [5] J. C. Ragusa and V. S. Mahadevan, “Consistent and accurate schemes for coupled neutronics thermal-hydraulics reactor analysis,” *Nuclear Engineering and Design*, vol. 239, no. 3, pp. 566–579, 2009.
- [6] D. Y. Anistratov and J. S. Warsa, “Discontinuous finite element quasi-diffusion methods,” *Nuclear Science and Engineering*, vol. 191, no. 2, pp. 105–120, 2018.

Appendices

Appendix A LD Discretization of Neutron Transport equation

The quasi-steady state transport equation for group g obtained using the backward Euler temporal discretization is

$$\mu \frac{\partial}{\partial x} \psi_g(x, \mu) + \hat{\sigma}_{t,g} \psi_g(x, \mu) = q_g(x, \mu) \quad x \in [0, X], \mu \in [-1, 1], g = 1, \dots, N_g \quad (52a)$$

$$\psi(0, \mu) = \psi^+, \quad \mu > 0 \quad (52b)$$

$$\psi(X, \mu) = \psi^-, \quad \mu < 0 \quad (52c)$$

where the time index has been suppressed, and

$$q_g(x, \mu) = \frac{\chi_g}{2} \sum_{g'=1}^{N_g} (1 - \beta_{g'}) \nu \sigma_{f,g'}(x) \phi_{g'}(x) + \frac{1}{2} \sum_{g'=1}^{N_g} (\sigma_{s,g' \rightarrow g}(x) + \Gamma_{g' \rightarrow g} \nu \sigma_{f,g'}) \phi_{g'}(x) + \hat{q}_g^{(j+1)}(x, \mu) + \bar{q}_{del,g}(x, \mu) \quad (53)$$

Here $\hat{\sigma}_{t,g}$ and $\hat{q}$ are the augmented cross sections from the time discretization, and $\hat{\sigma}_{s,g' \rightarrow g}$ is the effective scattering cross section from precursor decay.

Then let $\mathbb{V}_h$ be the space of spatially piecewise linear functions on grid $\{\kappa_i\}_{i=1}^{N_i}$, where

$$\kappa_i = [x_{i-1/2}, x_{i+1/2}], \quad (54)$$

$$x_{1/2} = 0, \quad (55)$$

$$x_{N_i-1/2} = X. \quad (56)$$

Then let ψ, ϕ , and q lie in $\mathbb{V}_h$ and suppose all coefficients are constant within each cell (note that this requires cross section evaluation using average temperature in each cell). Each function can be represented by the basis functions in each cell

$$b_{L,i} = \frac{x_{i+1/2} - x}{\Delta x_i}, \quad b_{R,i} = \frac{x - x_{i-1/2}}{\Delta x_i}, \quad (57)$$

Integrating over each basis function in each cell, applying the divergence theorem to the streaming term, and using the upwind flux defined by

$$\psi_{g,L,i}^b = \begin{cases} \psi_{g,R,i-1}, & \mu > 0 \\ \psi_{g,L,i}, & \mu < 0 \end{cases}, \quad \psi_{g,R,i}^b = \begin{cases} \psi_{g,R,i}, & \mu > 0 \\ \psi_{g,L,i+1}, & \mu < 0 \end{cases}, \quad (58)$$

the interior-cell equations are finally

$$\mu \left[-\psi_{g,L,i}^b \right] + \frac{1}{2} \mu (\psi_{g,L,i} + \psi_{g,R,i}) + \hat{\sigma}_{t,g} \frac{1}{6} \Delta x_i (2\psi_{g,L,i} + \psi_{g,R,i}) = \frac{1}{6} \Delta x_i (2q_{g,L,i} + q_{g,R,i}) \quad (59a)$$

$$\mu \left[\psi_{g,R,i}^b \right] - \frac{1}{2} \mu (\psi_{g,L,i} + \psi_{g,R,i}) + \hat{\sigma}_{t,g} \frac{1}{6} \Delta x_i (\psi_{g,L,i} + 2\psi_{g,R,i}) = \frac{1}{6} \Delta x_i (q_{g,L,i} + 2q_{g,R,i}) \quad (59b)$$

for $i = 2, \dots, N_x - 1$, where μ dependence of ψ and q have been suppressed. Applying the boundary conditions gives the equations for the left face and $\mu > 0$ directions,

$$\frac{1}{2} \mu (\psi_{g,L,i} + \psi_{g,R,i}) + \hat{\sigma}_{t,g} \frac{1}{6} \Delta x_i (2\psi_{g,L,i} + \psi_{g,R,i}) = \frac{1}{6} \Delta x_i (2q_{g,L,i} + q_{g,R,i}) + \mu \psi_g^+ \quad (60a)$$

$$\mu \left[\psi_{g,R,i}^b \right] - \frac{1}{2} \mu (\psi_{g,L,i} + \psi_{g,R,i}) + \hat{\sigma}_{t,g} \frac{1}{6} \Delta x_i (\psi_{g,L,i} + 2\psi_{g,R,i}) = \frac{1}{6} \Delta x_i (q_{g,L,i} + 2q_{g,R,i}) + \mu \psi_g^+ \quad (60b)$$

$i = 1, \mu > 0$

and for the right face with $\mu < 0$,

$$\mu \left[-\psi_{g,L,i}^b \right] + \frac{1}{2} \mu (\psi_{g,L,i} + \psi_{g,R,i}) + \hat{\sigma}_{t,g} \frac{1}{6} \Delta x_i (2\psi_{g,L,i} + \psi_{g,R,i}) = \frac{1}{6} \Delta x_i (2q_{g,L,i} + q_{g,R,i}) - \mu \psi_g^- \quad (61a)$$

$$-\frac{1}{2} \mu (\psi_{g,L,i} + \psi_{g,R,i}) + \hat{\sigma}_{t,g} \frac{1}{6} \Delta x_i (\psi_{g,L,i} + 2\psi_{g,R,i}) = \frac{1}{6} \Delta x_i (q_{g,L,i} + 2q_{g,R,i}) - \mu \psi_g^- \quad (61b)$$

$i = N_x, \mu < 0$

Appendix B LD Discretization of QD equations

The LD discretization of the QD equations follows the method outlined in [6]. The unclosed zeroth and first moment equations are, for group g (using the same terms as eq. (16)),

$$\frac{\partial}{\partial x} J_g(x) + \sigma_{t,g} \phi_g(x) = \chi_g \overline{(1-\beta)\nu\sigma_f} \phi + \sum_{g'=1}^{N_g} \sigma_{g' \rightarrow g} \phi_{g'}(x) + Q_g(x) \quad (62a)$$

$$\frac{\partial}{\partial x} F(x) + \sigma_{t,g} J_g = q_g \quad (62b)$$

$$J_g(0) = C_{0,g}^b \phi_g(0), \quad J_g(X) = C_{X,g}^b \phi_g(X) \quad (62c)$$

where here $\sigma_{g' \rightarrow g}$ represents the total transfer cross section including both scattering, Q and q represent precursor decay, indepdent source, and source from the previous time step, and F is the second angular moment,

$$F(x) = \int_{-1}^1 \mu^2 \psi(x, \mu) d\mu \quad (63)$$

We apply the same spatial grid $\{\kappa_i\}_{i=1}^{N_x}$ as for the high-order transport equation, and suppose all unknowns and sources belong to the same space $\mathbb{V}_h$;

$$\phi_{g,i}(x) = \sum_{c=L,R} \phi_{g,i,c} b_{i,c}(x) \quad (64)$$

$$J_{g,i}(x) = \sum_{c=L,R} J_{g,i,c} b_{i,c}(x) \quad (65)$$

Multiplying eqs. (62a) and (62b) by the basis functions and integrating over each cell (and applying the divergence theorem on spatial derivative terms) yields

$$\begin{aligned} -J_g^b(x_{i-1/2}) + \frac{1}{2} (J_{g,i,L} + J_{g,i,R}) + \sigma_{t,g} \frac{\Delta x}{6} (2\phi_{g,i,L} + \phi_{g,i,R}) &= \chi_g \overline{(1-\beta)\nu\sigma_f} \frac{\Delta x}{6} (2\phi_{i,L} + \phi_{i,R}) \\ &+ \sum_{g'=1}^{N_g} \sigma_{g' \rightarrow g} \frac{\Delta x}{6} (2\phi_{g',i,L} + \phi_{g',i,R}) + \frac{\Delta x}{6} (2Q_{g,i,L} + Q_{g,i,R}) \end{aligned} \quad (66a)$$

$$\begin{aligned} J_g^b(x_{i+1/2}) - \frac{1}{2} (J_{g,i,L} + J_{g,i,R}) + \sigma_{t,g} \frac{\Delta x}{6} (\phi_{g,i,L} + 2\phi_{g,i,R}) &= \chi_g \overline{(1-\beta)\nu\sigma_f} \frac{\Delta x}{6} (\phi_{i,L} + 2\phi_{i,R}) \\ &+ \sum_{g'=1}^{N_g} \sigma_{g' \rightarrow g} \frac{\Delta x}{6} (\phi_{g',i,L} + 2\phi_{g',i,R}) + \frac{\Delta x}{6} (Q_{g,i,L} + 2Q_{g,i,R}) \end{aligned} \quad (66b)$$

$$-F_g^b(x_{i-1/2}) + \frac{1}{\Delta x} \int_{\kappa_i} F_i(x) dx + \sigma_{t,g} \frac{\Delta x}{6} (2J_{g,i,L} + J_{g,i,R}) = \frac{\Delta x}{6} (2q_{g,i,L} + q_{g,i,R}) \quad (66c)$$

$$F_g^b(x_{i+1/2}) - \frac{1}{\Delta x} \int_{\kappa_i} F_i(x) dx + \sigma_{t,g} \frac{\Delta x}{6} (J_{g,i,L} + 2J_{g,i,R}) = \frac{\Delta x}{6} (q_{g,i,L} + 2q_{g,i,R}) \quad (66d)$$

where the mesh is uniform and spatial index has been suppressed from coefficients (the cross sections must be constant in each cell).

The equations are closed by defining the integrated second moment as

$$\int_{\kappa_i} F_i(x) dx = \bar{E}_{g,i} \int_{\kappa_i} \phi_{g,i}(x) dx \quad (67)$$

where the cell-averaged Eddington factor is

$$\bar{E}_{g,i} = \frac{\int_{\kappa_i} dx \int_{-1}^1 d\mu \mu^2 \psi_{g,i}(x, \mu)}{\int_{\kappa_i} dx \int_{-1}^1 d\mu \psi_{g,i}(x, \mu)}. \quad (68)$$

The second moment at the boundary is defined with

$$F_g^b(x_{i+1/2}) = E_{g,i+1/2}^b \phi_{g,i+1/2}^b \quad (69)$$

where the cell-edge Eddington factor is

$$E_{g,i+1/2}^b = \frac{\int_{-1}^0 \mu^2 \psi_{g,i+1}(x_{i+1/2}, \mu) d\mu + \int_0^1 \mu^2 \psi_{g,i}(x_{i+1/2}, \mu) d\mu}{\int_{-1}^0 \psi_{g,i+1}(x_{i+1/2}, \mu) d\mu + \int_0^1 \psi_{g,i}(x_{i+1/2}, \mu) d\mu}. \quad (70)$$

The Miften-Larsen interface conditions [6] are used,

$$\begin{aligned} \phi_{g,i+1/2}^b &= \phi_{g,i}^+(x_{i+1/2}) + \phi_{g,i+1}^-(x_{i+1/2}) \\ &= \frac{1}{C_{g,i}^+(x_{i+1/2})} (J_{g,i,R} + G_{g,i}(x_{i+1/2}) \phi_{g,i,R}) \\ &\quad + \frac{1}{C_{g,i+1}^-(x_{i+1/2})} (J_{g,i+1,L} + G_{g,i+1}(x_{i+1/2}) \phi_{g,i+1,L}) \end{aligned} \quad (71a)$$

$$\begin{aligned} J_{g,i+1/2}^b &= J_{g,i}^+(x_{i+1/2}) + J_{g,i+1}^-(x_{i+1/2}) \\ &= \frac{1}{2} (J_{g,i,R} + J_{g,i+1,L} + G_{g,i}(x_{i+1/2}) \phi_{g,i,R} - G_{g,i+1}(x_{i+1/2}) \phi_{g,i+1,L}) \end{aligned} \quad (71b)$$

where

$$G_{g,i}(x_{i+1/2}) = \frac{\int_{-1}^1 |\mu| \psi_{g,i,R}(\mu) d\mu}{\int_{-1}^1 \psi_{g,i,R}(\mu) d\mu} \quad (72)$$

$$C_{g,i}^+(x_{i+1/2}) = \frac{\int_0^1 \mu \psi_{g,i,R}(\mu) d\mu}{\int_0^1 \psi_{g,i,R}(\mu) d\mu} \quad (73)$$

$$C_{g,i}^+(x_{i+1/2}) = \frac{\int_{-1}^0 \mu \psi_{g,i,R}(\mu) d\mu}{\int_{-1}^0 \psi_{g,i,R}(\mu) d\mu} \quad (74)$$

Then eq. (66) becomes, using these closure and interface terms,

$$\begin{aligned} & - \left[\frac{1}{2} (J_{g,i-1,R} + J_{g,i,L} + G_{g,i-1}(x_{i-1/2})\phi_{g,i-1,R} - G_{g,i}(x_{i-1/2})\phi_{g,i,L}) \right] \\ & + \frac{1}{2} (J_{g,i,L} + J_{g,i,R}) + \sigma_{t,g} \frac{\Delta x}{6} (2\phi_{g,i,L} + \phi_{g,i,R}) = \chi_g \overline{(1-\beta)\nu\sigma_f} \frac{\Delta x}{6} (2\phi_{i,L} + \phi_{i,R}) \\ & + \sum_{g'=1}^{N_g} \sigma_{g' \rightarrow g} \frac{\Delta x}{6} (2\phi_{g',i,L} + \phi_{g',i,R}) + \frac{\Delta x}{6} (2Q_{g,i,L} + Q_{g,i,R}) \quad (75a) \end{aligned}$$

$$\begin{aligned} & \left[\frac{1}{2} (J_{g,i,R} + J_{g,i+1,L} + G_{g,i}(x_{i+1/2})\phi_{g,i,R} - G_{g,i+1}(x_{i+1/2})\phi_{g,i+1,L}) \right] \\ & - \frac{1}{2} (J_{g,i,L} + J_{g,i,R}) + \sigma_{t,g} \frac{\Delta x}{6} (\phi_{g,i,L} + 2\phi_{g,i,R}) = \chi_g \overline{(1-\beta)\nu\sigma_f} \frac{\Delta x}{6} (\phi_{i,L} + 2\phi_{i,R}) \\ & + \sum_{g'=1}^{N_g} \sigma_{g' \rightarrow g} \frac{\Delta x}{6} (\phi_{g',i,L} + 2\phi_{g',i,R}) + \frac{\Delta x}{6} (Q_{g,i,L} + 2Q_{g,i,R}) \quad (75b) \end{aligned}$$

$$\begin{aligned} & - E_{g,i-1/2}^b \left[\frac{1}{C_{g,i-1}^+(x_{i-1/2})} (J_{g,i-1,R} + G_{g,i-1}(x_{i-1/2})\phi_{g,i-1,R}) \right. \\ & \quad \left. + \frac{1}{C_{g,i}^-(x_{i-1/2})} (J_{g,i,L} + G_{g,i}(x_{i-1/2})\phi_{g,i,L}) \right] + \frac{1}{2} \bar{E}_{g,i} (\phi_{g,i,L} + \phi_{g,i,R}) \\ & \quad + \sigma_{t,g} \frac{\Delta x}{6} (2J_{g,i,L} + J_{g,i,R}) = \frac{\Delta x}{6} (2q_{g,i,L} + q_{g,i,R}) \quad (75c) \end{aligned}$$

$$\begin{aligned} & E_{g,i+1/2}^b \left[\frac{1}{C_{g,i}^+(x_{i+1/2})} (J_{g,i,R} + G_{g,i}(x_{i+1/2})\phi_{g,i,R}) \right. \\ & \quad \left. + \frac{1}{C_{g,i+1}^-(x_{i+1/2})} (J_{g,i+1,L} + G_{g,i+1}(x_{i+1/2})\phi_{g,i+1,L}) \right] - \frac{1}{2} \bar{E}_{g,i} (\phi_{g,i,L} + \phi_{g,i,R}) \\ & \quad + \sigma_{t,g} \frac{\Delta x}{6} (J_{g,i,L} + 2J_{g,i,R}) = \frac{\Delta x}{6} (q_{g,i,L} + 2q_{g,i,R}) \\ & \quad i = 2, \dots, N_x - 1 \quad (75d) \end{aligned}$$

On the leftmost cell, where with vacuum boundary conditions $\phi^+(0) = J^+(0) = 0$, eqs. (75a) and (75c) are replaced with

$$\begin{aligned}
& - \left[\frac{1}{2} (J_{g,i,L} - G_{g,i}(x_{i-1/2})\phi_{g,i,L}) \right] \\
& + \frac{1}{2} (J_{g,i,L} + J_{g,i,R}) + \sigma_{t,g} \frac{\Delta x}{6} (2\phi_{g,i,L} + \phi_{g,i,R}) = \chi_g \overline{(1-\beta)\nu\sigma_f} \frac{\Delta x}{6} (2\phi_{i,L} + \phi_{i,R}) \\
& + \sum_{g'=1}^{N_g} \sigma_{g' \rightarrow g} \frac{\Delta x}{6} (2\phi_{g',i,L} + \phi_{g',i,R}) + \frac{\Delta x}{6} (2Q_{g,i,L} + Q_{g,i,R}) \quad (76a)
\end{aligned}$$

$$\begin{aligned}
& - E_{g,0}^b \left[\frac{1}{C_{g,i}^-(x_{i-1/2})} (J_{g,i,L} + G_{g,i}(x_{i-1/2})\phi_{g,i,L}) \right] + \frac{1}{2} \bar{E}_{g,i} (\phi_{g,i,L} + \phi_{g,i,R}) \\
& + \sigma_{t,g} \frac{\Delta x}{6} (2J_{g,i,L} + J_{g,i,R}) = \frac{\Delta x}{6} (2q_{g,i,L} + q_{g,i,R}), \quad i = 1 \quad (76b)
\end{aligned}$$

where

$$E_{g,0}^b = \frac{\int_{-1}^0 \mu^2 \psi_{g,i}(x_{i-1/2}, \mu) d\mu}{\int_{-1}^0 \psi_{g,i}(x_{i-1/2}, \mu) d\mu}, \quad i = 1 \quad (77)$$

and on the rightmost cell, where $\phi^-(X) = J^-(X) = 0$, eqs. (75b) and (75d) become

$$\begin{aligned}
& \left[\frac{1}{2} (J_{g,i,R} + G_{g,i}(x_{i+1/2})\phi_{g,i,R}) \right] \\
& - \frac{1}{2} (J_{g,i,L} + J_{g,i,R}) + \sigma_{t,g} \frac{\Delta x}{6} (\phi_{g,i,L} + 2\phi_{g,i,R}) = \chi_g \overline{(1-\beta)\nu\sigma_f} \frac{\Delta x}{6} (\phi_{i,L} + 2\phi_{i,R}) \\
& + \sum_{g'=1}^{N_g} \sigma_{g' \rightarrow g} \frac{\Delta x}{6} (\phi_{g',i,L} + 2\phi_{g',i,R}) + \frac{\Delta x}{6} (Q_{g,i,L} + 2Q_{g,i,R}) \quad (78a)
\end{aligned}$$

$$\begin{aligned}
& E_{g,N_x+1/2}^b \left[\frac{1}{C_{g,i}^+(x_{i+1/2})} (J_{g,i,R} + G_{g,i}(x_{i+1/2})\phi_{g,i,R}) \right] - \frac{1}{2} \bar{E}_{g,i} (\phi_{g,i,L} + \phi_{g,i,R}) \\
& + \sigma_{t,g} \frac{\Delta x}{6} (J_{g,i,L} + 2J_{g,i,R}) = \frac{\Delta x}{6} (q_{g,i,L} + 2q_{g,i,R}) \quad i = 2, \dots, N_x - 1 \quad (78b)
\end{aligned}$$

where

$$E_{g,N_x+1/2}^b = \frac{\int_0^1 \mu^2 \psi_{g,i}(x_{i+1/2}, \mu) d\mu}{\int_0^1 \psi_{g,i}(x_{i+1/2}, \mu) d\mu}, \quad i = N_x. \quad (79)$$

Equations (75), (76) and (78) give the equations for the LDFEM discretization of the QD equations with vacuum boundaries, cellwise constant coefficients, and Miften-Larsen interface conditions.

Appendix C LDFEM Discretization of advection-diffusion equation

The advection-diffusion equation in 1 dimension is

$$\left[r - \frac{\partial}{\partial x} k_b \frac{\partial}{\partial x} + \dot{m} c_p \frac{\partial}{\partial x} \right] T^{(j+1)}(x) = Q(x) \quad (80)$$

where r is the local removal term from the backward Euler temporal discretization. For this model, k_b and $\dot{m} c_p$ will be spatially constant within each cell, $Q(x)$ will be piecewise linear, r will be constant everywhere, and the first derivative of k_b will be ignored. For this problem the boundary conditions

$$T(0) = T^{in}, \quad \left. \frac{dT}{dx} \right|_{x=X} = 0 \quad (81)$$

will be used. Then let $\mathbb{V}_h$ be the space of spatially piecewise linear functions on grid $\{\kappa_i\}_{i=1}^{N_i}$, where

$$\kappa_i = [x_{i-1/2}, x_{i+1/2}], \quad (82)$$

$$x_{1/2} = 0, \quad (83)$$

$$x_{N_i-1/2} = X \quad (84)$$

with cell width Δx_i and cell center x_i .

Let T and Q lie in $\mathbb{V}_h$ and suppose all coefficients are constant within each cell. Each function can be represented by the basis functions in each cell using the linear combination

$$T_i(x) = \bar{T}_i b_i^z(x) + \hat{T}_i b_i^f(x) \quad (85)$$

where

$$b_i^z(x) = 1, \quad b_i^f(x) = \frac{2}{\Delta x_i}(x - x_i) \quad (86)$$

where

$$\int_{\kappa_i} b_i^z(x) dx = \Delta x_i \quad (87)$$

$$\int_{\kappa_i} b_i^f(x) dx = 0 \quad (88)$$

$$\int_{\kappa_i} (b_i^z(x))^2 dx = \Delta x_i \quad (89)$$

$$\int_{\kappa_i} (b_i^f(x))^2 dx = \frac{1}{3} \Delta x_i \quad (90)$$

$$\int_{\kappa_i} b_i^z(x) b_i^f(x) dx = 0 \quad (91)$$

Here $\bar{T}_i$ represents the average value of T in κ_i and $\hat{T}_i$ is defined such that the corner values are $\bar{T} \pm \hat{T}$. The advection-diffusion equation is multiplied by each basis function and integrated over the cell,

$$r_i \int_{\kappa_i} T(x) b^d(x) dx - \int_{\kappa_i} \frac{\partial}{\partial x} k_{b,i} \frac{\partial T}{\partial x} b^d(x) dx + (\dot{m}c_p)_i \int_{\kappa_i} \frac{\partial T}{\partial x} b^d(x) dx = \int_{\kappa_i} Q(x) b^d(x) dx \quad (92)$$

Integration by parts is applied to move the derivative terms onto the basis functions,

$$\begin{aligned} r_i \int_{\kappa_i} T(x) b_i^d(x) dx - k_i \left[b_i^d(x) \frac{\partial T(x)}{\partial x} \Big|_{x_{i-1/2}}^{x_{i+1/2}} - \int_{\kappa_i} \frac{\partial}{\partial x} T(x) \cdot \frac{\partial}{\partial x} b_i^d(x) dx \right] \\ + (\dot{m}c_p)_i \left[T(x) b_i^d(x) \Big|_{x_{i-1/2}}^{x_{i+1/2}} - \int_{\kappa_i} T(x) \frac{\partial}{\partial x} b_i^d(x) dx \right] = \int_{\kappa_i} Q(x) b^d(x) dx \end{aligned} \quad (93)$$

once more integration by parts is applied to the diffusion term,

$$\begin{aligned} r_i \int_{\kappa_i} T(x) b_i^d(x) dx - k_i \left[b_i^d(x) \frac{\partial T(x)}{\partial x} \Big|_{x_{i-1/2}}^{x_{i+1/2}} - T(x) \frac{\partial}{\partial x} b_i^d(x) \Big|_{x_{i-1/2}}^{x_{i+1/2}} \right] \\ + (\dot{m}c_p)_i \left[T(x) b_i^d(x) \Big|_{x_{i-1/2}}^{x_{i+1/2}} - \int_{\kappa_i} T(x) \frac{\partial}{\partial x} b_i^d(x) dx \right] = \int_{\kappa_i} Q(x) b^d(x) dx. \end{aligned} \quad (94)$$

For the diffusion term the interface condition is applied,

$$\frac{dT}{dx} \Big|_{x_{i-1/2}} = \frac{\hat{T}_i}{\Delta x_i} + \frac{\hat{T}_{i-1}}{\Delta x_{i-1}} \quad (95a)$$

$$\frac{dT}{dx} \Big|_{x_{i+1/2}} = \frac{\hat{T}_i}{\Delta x_i} + \frac{\hat{T}_{i+1}}{\Delta x_{i+1}} \quad (95b)$$

Then this is evaluated with $d = z, f$,

$$r_i \bar{T}_i \Delta x_i + k_i \left(\frac{\hat{T}_{i-1}}{\Delta x_{i-1}} - \frac{\hat{T}_{i+1}}{\Delta x_{i+1}} \right) + (\dot{m}c_p)_i T(x) \Big|_{x_{i-1/2}}^{x_{i+1/2}} = \Delta x_i \bar{Q}_i \quad (96a)$$

$$\begin{aligned} \frac{1}{3} r_i \hat{T}_i \Delta x_i - k_i \left[\frac{\hat{T}_{i+1}}{\Delta x_{i+1}} + 2 \frac{\hat{T}_i}{\Delta x_i} + \frac{\hat{T}_{i-1}}{\Delta x_{i-1}} - \frac{2}{\Delta x_i} T(x) \Big|_{x_{i-1/2}}^{x_{i+1/2}} \right] \\ + (\dot{m}c_p)_i [T(x_{i+1/2}) + T(x_{i-1/2}) - 2\bar{T}_i] = \frac{1}{3} \hat{Q}_i \Delta x_i \end{aligned} \quad (96b)$$

The interface condition for T depends on the physics of the problem.

C.1 Advective Interface Condition

In a strongly advective problem, the interface condition is

$$T(x_{i+1/2}) = \bar{T}_i + \hat{T}_i, \quad T(x_{i-1/2}) = \bar{T}_{i-1} + \hat{T}_{i-1}, \quad (97)$$

and thus the interior equations become

$$r_i \bar{T}_i \Delta x_i + k_i \left(\frac{\hat{T}_{i-1}}{\Delta x_{i-1}} - \frac{\hat{T}_{i+1}}{\Delta x_{i+1}} \right) + (\dot{m}c_p)_i (\bar{T}_i + \hat{T}_i - \bar{T}_{i-1} - \hat{T}_{i-1}) = \Delta x_i \bar{Q}_i \quad (98a)$$

$$\begin{aligned} \frac{1}{3} r_i \hat{T}_i \Delta x_i + k_i \left[-\frac{\hat{T}_{i+1}}{\Delta x_{i+1}} - \frac{\hat{T}_{i-1}}{\Delta x_{i-1}} + \frac{2}{\Delta x_i} (\bar{T}_i - \bar{T}_{i-1} - \hat{T}_{i-1}) \right] \\ + (\dot{m}c_p)_i [-\bar{T}_i + \hat{T}_i + \bar{T}_{i-1} + \hat{T}_{i-1}] = \frac{1}{3} \hat{Q}_i \Delta x_i \end{aligned} \quad (98b)$$

With a fixed-temperature boundary condition at the left face, this is

$$r_i \bar{T}_i \Delta x_i + k_i \left(\frac{\hat{T}_{i-1}}{\Delta x_{i-1}} - \frac{\hat{T}_{i+1}}{\Delta x_{i+1}} \right) + (\dot{m}c_p)_i (\bar{T}_i + \hat{T}_i) = \Delta x_i \bar{Q}_i + (\dot{m}c_p)_i T_{in}^L \quad (99a)$$

$$\begin{aligned} \frac{1}{3} r_i \hat{T}_i \Delta x_i + k_i \left[-\frac{\hat{T}_{i+1}}{\Delta x_{i+1}} - 2\frac{\hat{T}_i}{\Delta x_i} - \frac{\hat{T}_{i-1}}{\Delta x_{i-1}} + \frac{2}{\Delta x_i} (\bar{T}_i + \hat{T}_i) \right] \\ + (\dot{m}c_p)_i [\hat{T}_i - \bar{T}_i] = \frac{1}{3} \hat{Q}_i \Delta x_i + \left(2\frac{k_i}{\Delta x_i} - (\dot{m}c_p)_i \right) T_{in}^L \end{aligned} \quad (99b)$$

and, setting the first derivative to zero on the right face,

$$r_i \bar{T}_i \Delta x_i + k_i \left(-\frac{\hat{T}_{i-1}}{\Delta x_{i-1}} - \frac{\hat{T}_i}{\Delta x_i} \right) + (\dot{m}c_p)_i (\bar{T}_i + \hat{T}_i - \bar{T}_{i-1} - \hat{T}_{i-1}) = \Delta x_i \bar{Q}_i \quad (100a)$$

$$\begin{aligned} \frac{1}{3} r_i \hat{T}_i \Delta x_i + k_i \left[\frac{\hat{T}_i}{\Delta x_i} - \frac{\hat{T}_{i-1}}{\Delta x_{i-1}} + \frac{2}{\Delta x_i} (\bar{T}_i - \bar{T}_{i-1} - \hat{T}_{i-1}) \right] \\ + (\dot{m}c_p)_i [-\bar{T}_i + \hat{T}_i + \bar{T}_{i-1} + \hat{T}_{i-1}] = \frac{1}{3} \hat{Q}_i \Delta x_i \end{aligned} \quad (100b)$$

C.2 Diffusive Interface Condition

In a strongly (or purely) diffusive problem, the interface condition is

$$T(x_{i+1/2}) = \frac{1}{2} (\bar{T}_i + \hat{T}_i + \bar{T}_{i+1} - \hat{T}_{i+1}), \quad T(x_{i-1/2}) = \frac{1}{2} (\bar{T}_i - \hat{T}_i + \bar{T}_{i-1} + \hat{T}_{i-1}), \quad (101)$$

and thus the interior equations are

$$r_i \bar{T}_i \Delta x_i + k_i \left(\frac{\hat{T}_{i-1}}{\Delta x_{i-1}} - \frac{\hat{T}_{i+1}}{\Delta x_{i+1}} \right) + \frac{(\dot{m}c_p)_i}{2} (2\hat{T}_i + \bar{T}_{i+1} - \hat{T}_{i+1} - \bar{T}_{i-1} - \hat{T}_{i-1}) = \Delta x_i \bar{Q}_i \quad (102a)$$

$$\begin{aligned} \frac{1}{3} r_i \hat{T}_i \Delta x_i + k_i \left[-\frac{\hat{T}_{i+1}}{\Delta x_{i+1}} - 2\frac{\hat{T}_i}{\Delta x_i} - \frac{\hat{T}_{i-1}}{\Delta x_{i-1}} + \frac{1}{\Delta x_i} (2\hat{T}_i + \bar{T}_{i+1} - \hat{T}_{i+1} - \bar{T}_{i-1} - \hat{T}_{i-1}) \right] \\ + (\dot{m}c_p)_i \left[\frac{1}{2} (\bar{T}_{i+1} - \hat{T}_{i+1} + \bar{T}_{i-1} + \hat{T}_{i+1}) - \bar{T}_i \right] = \frac{1}{3} \hat{Q}_i \Delta x_i. \end{aligned} \quad (102b)$$

The fixed-temperature left-side boundary equation is

$$r_i \bar{T}_i \Delta x_i + k_i \left(-\frac{\hat{T}_{i+1}}{\Delta x_{i+1}} + \frac{\hat{T}_i}{\Delta x_i} \right) + \frac{(\dot{m}c_p)_i}{2} (\bar{T}_i + \hat{T}_i + \bar{T}_{i+1} - \hat{T}_{i+1}) = \Delta x_i \bar{Q}_i + (\dot{m}c_p)_i T_{in}^L \quad (103a)$$

$$\begin{aligned} \frac{1}{3} r_i \hat{T}_i \Delta x_i + k_i \left[-\frac{\hat{T}_{i+1}}{\Delta x_{i+1}} - 3\frac{\hat{T}_i}{\Delta x_i} + \frac{1}{\Delta x_i} (\bar{T}_i + \hat{T}_i + \bar{T}_{i+1} - \hat{T}_{i+1}) \right] \\ + (\dot{m}c_p)_i \left[-\frac{3}{2} \bar{T}_i + \frac{1}{2} (\hat{T}_i + \bar{T}_{i+1} - \hat{T}_{i+1}) \right] = \frac{1}{3} \hat{Q}_i \Delta x_i + \left(2\frac{k_i}{\Delta x_i} - (\dot{m}c_p)_i \right) T_{in}^L \end{aligned} \quad (103b)$$

and the fixed-temperature right-side boundary equation is

$$r_i \bar{T}_i \Delta x_i + k_i \left(-\frac{\hat{T}_i}{\Delta x_i} + \frac{\hat{T}_{i-1}}{\Delta x_{i-1}} \right) + \frac{(\dot{m}c_p)_i}{2} (-\bar{T}_i + \hat{T}_i - \bar{T}_{i-1} - \hat{T}_{i-1}) = \Delta x_i \bar{Q}_i + (\dot{m}c_p)_i T_{in}^R \quad (104a)$$

$$\begin{aligned} \frac{1}{3} r_i \hat{T}_i \Delta x_i + k_i \left[-3\frac{\hat{T}_i}{\Delta x_i} - \frac{\hat{T}_{i-1}}{\Delta x_{i-1}} + \frac{1}{\Delta x_i} (-\bar{T}_i + \hat{T}_i - \bar{T}_{i-1} - \hat{T}_{i-1}) \right] \\ + (\dot{m}c_p)_i \left[\frac{1}{2} (-\bar{T}_i + \hat{T}_i + \bar{T}_{i-1} + \hat{T}_{i-1}) \right] = \frac{1}{3} \hat{Q}_i \Delta x_i - \left(2\frac{k_i}{\Delta x_i} + (\dot{m}c_p)_i \right) T_{in}^R \end{aligned} \quad (104b)$$

The insulated left-side boundary equation is

$$r_i \bar{T}_i \Delta x_i + k_i \left(-\frac{\hat{T}_i}{\Delta x_i} - \frac{\hat{T}_{i+1}}{\Delta x_{i+1}} \right) + \frac{(\dot{m}c_p)_i}{2} (-\bar{T}_i + 3\hat{T}_i + \bar{T}_{i+1} - \hat{T}_{i+1}) = \Delta x_i \bar{Q}_i \quad (105a)$$

$$\begin{aligned} \frac{1}{3} r_i \hat{T}_i \Delta x_i + k_i \left[-\frac{\hat{T}_i}{\Delta x_i} - \frac{\hat{T}_{i+1}}{\Delta x_{i+1}} + \frac{1}{\Delta x_i} (-\bar{T}_i + 3\hat{T}_i + \bar{T}_{i+1} - \hat{T}_{i+1}) \right] \\ + \frac{(\dot{m}c_p)_i}{2} [-\bar{T}_i - \hat{T}_i + \bar{T}_{i+1} - \hat{T}_{i+1}] = \frac{1}{3} \hat{Q}_i \Delta x_i. \end{aligned} \quad (105b)$$

and the insulated right-side boundary equation is

$$r_i \bar{T}_i \Delta x_i + k_i \left(\frac{\hat{T}_i}{\Delta x_i} + \frac{\hat{T}_{i-1}}{\Delta x_{i-1}} \right) + \frac{(\dot{m} c_p)_i}{2} (\bar{T}_i + 3\hat{T}_i - \bar{T}_{i-1} - \hat{T}_{i-1}) = \Delta x_i \bar{Q}_i \quad (106a)$$

$$\begin{aligned} \frac{1}{3} r_i \hat{T}_i \Delta x_i + k_i \left[-\frac{\hat{T}_i}{\Delta x_i} - \frac{\hat{T}_{i-1}}{\Delta x_{i-1}} + \frac{1}{\Delta x_i} (\bar{T}_i + 3\hat{T}_i - \bar{T}_{i-1} - \hat{T}_{i-1}) \right] \\ + \frac{(\dot{m} c_p)_i}{2} [-\bar{T}_i + \hat{T}_i + \bar{T}_{i-1} + \hat{T}_{i-1}] = \frac{1}{3} \hat{Q}_i \Delta x_i. \end{aligned} \quad (106b)$$

Appendix D Computational Units

The original system of time-dependent equations for the multiphysics problem is

$$\begin{aligned} \frac{1}{v_g} \frac{\partial}{\partial t} \psi + \mu \frac{\partial}{\partial x} \psi + \sigma_t \psi = \frac{1}{2} \int_0^\infty dE' \sigma_s(E' \rightarrow E) \int_{-1}^1 d\mu' \psi(\mu') \\ + \frac{\chi(E)}{2} \int_0^\infty dE' \nu \sigma_f(E') \int_{-1}^1 d\mu' \psi(\mu') + \frac{1}{2} \sum_k \chi_k^d(E) \lambda_k C_k + q \end{aligned} \quad (107a)$$

$$\phi(x, E) = \int_{-1}^1 d\mu \psi(x, \mu, E) \quad (107b)$$

$$\left[\frac{\partial}{\partial t} - \frac{\partial}{\partial x} k_b \frac{\partial}{\partial x} + \dot{m} c_p \frac{\partial}{\partial x} \right] T(x) = W_f \int_0^\infty dE \nu \sigma_f(E) \phi(x, E) \quad (107c)$$

$$\sigma_u(x, E) = \sigma_u(E, T(x), f(x)) \quad (107d)$$

$$\frac{\partial}{\partial t} C_k = -\lambda_k C_k + \int_0^\infty \beta_k(E) \nu \sigma_f(x, E) \phi(x, E) \quad (107e)$$

Let the computational variables be

$$v_g = c_v \mathbf{v}_g \quad (108)$$

$$\psi = c_\psi \Psi_g \quad (109)$$

$$\phi = c_\phi \Phi_g \quad (110)$$

$$C_k = c_\psi \mathbf{C}_k \quad (111)$$

Then eq. (107) becomes

$$\begin{aligned} \frac{1}{c_v \mathbf{v}_g} \frac{\partial}{\partial t} \Psi + \mu \frac{\partial}{\partial x} \Psi + \sigma_t \Psi = & \frac{1}{2} \int_0^\infty dE' \sigma_s(E' \rightarrow E) \int_{-1}^1 d\mu' \Psi(\mu') \\ & + \frac{\chi(E)}{2} \int_0^\infty dE' \nu \sigma_f(E') \int_{-1}^1 d\mu' \Psi(\mu') + \frac{1}{2} \sum_k \chi_k^d(E) \lambda_k \mathbf{C}_k + \frac{q}{c_\psi} \end{aligned} \quad (112a)$$

$$\Phi(x, E) = \int_{-1}^1 d\mu \Phi(x, \mu, E) \quad (112b)$$

$$\left[\frac{\partial}{\partial t} - \frac{\partial}{\partial x} k_b \frac{\partial}{\partial x} + \dot{m} c_p \frac{\partial}{\partial x} \right] T(x) = \frac{W_f}{c_\psi} \int_0^\infty dE \nu \sigma_f(E) \Phi(x, E) \quad (112c)$$

$$\sigma_u(x, E) = \sigma_u(E, T(x), f(x)) \quad (112d)$$

$$\frac{\partial}{\partial t} \mathbf{C}_k = -\lambda_k \mathbf{C}_k + \int_0^\infty \beta_k(E) \nu \sigma_f(x, E) \Phi(x, E) \quad (112e)$$

where

$$f(x) = \frac{1}{c_\psi} \int_0^\infty dE \sigma_f(E) \Phi(E) \quad (113)$$

Appendix E List of Symbols

- j = time index
- m = angular index for method of discrete ordinates
- k = delayed neutron precursor group
- g = neutron energy group
- n, s, l, r = iterative index: outermost, $\dots$, innermost
- i = spatial cell index
- u = reaction index for cross sections

TODO physics constants, etc